F18CF - Assignment 2

Instructions:

- This assignment is due at 23:59 (Edinburgh Time) on the date specified above
- Your solutions can be written or typeset. No paper submission is necessary
- *Gradescope* submission link available on assignment page in Canvas
- The assessment criteria are listed on the assignment page
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 4 questions worth a total of 40 marks.
- Marks are listed with each question.
- An additional 2 mark is awarded for meeting the submission standards.

Questions:

9 marks

1. A hospital tracks six clinical indicators for each patient visit. Stacking four visits as rows and six indicators as columns gives the data matrix:

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 2 & 4 & 3 & 7 & 7 & 4 \\ 1 & 2 & 2 & 5 & 5 & 6 \\ 3 & 6 & 6 & 15 & 14 & 15 \end{pmatrix}.$$

(a) For $k = 1, \ldots, 6$, define M_k as the submatrix of A consisting of the first k columns $C_1, C_2, \ldots, C_k$. Find the ranks of M_k for $k = 1, \ldots, 6$.

Solution:

Definition 4.42 tells us the rank is the dimension of the image, which is found by the number of linearly independent vectors in the span of the columns (Defn 3.40). We start with $M_6 = A$.

$$A = \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 2 & 4 & 3 & 7 & 7 & 4 \\ 1 & 2 & 2 & 5 & 5 & 6 \\ 3 & 6 & 6 & 15 & 14 & 15 \end{pmatrix} \sim \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 1 & 3 & 2 & 5 \\ 0 & 0 & 3 & 9 & 5 & 12 \end{pmatrix} \begin{matrix} R_2 \to R_2 - 2R_1 \\ R_3 \to R_3 - R_1 \\ R_4 \to R_4 - 3R_1 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 3 & 9 & 5 & 12 \end{pmatrix} \quad R_3 \to R_3 - R_2$$

$$\sim \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 6 \end{pmatrix} \quad R_4 \to R_4 - 3R_2$$

$$\sim \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad R_4 \to R_4 - 2R_3$$

We note this series of EROs will give us the REF for each M_k. From this, we have:

$$M_1 = [C_1] \sim \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad M_4 = [C_1|C_2|C_3|C_4] \sim \begin{pmatrix} 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$M_2 = [C_1|C_2] \sim \begin{pmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} \qquad M_5 = [C_1|C_2|C_3|C_4|C_5] \sim \begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

$$M_3 = [C_1|C_2|C_3] \sim \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad M_6 = [C_1|C_2|C_3|C_4|C_5|C_6] \sim \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

From the reduction above (pivots in columns $1, 3, 5$), we obtain: $\text{rank}(M_1) = 1$, $\text{rank}(M_2) = 1$, $\text{rank}(M_3) = 2$, $\text{rank}(M_4) = 2$, $\text{rank}(M_5) = 3$, and $\text{rank}(M_6) = 3$.

k	1	2	3	4	5	6
$\text{rank}(M_k)$	1	1	2	2	3	3

(b) Which columns C_{k+1} are linear combinations of the preceding columns $C_1, \ldots, C_k$?

Solution: Consider the vector equation

$$x_1C_1 + x_2C_2 + \cdots + x_kC_k \;=\; C_{k+1}.$$

This yields a linear system with coefficient matrix $M_k = [C_1|\cdots|C_k]$ and augmented matrix $[M_k|C_{k+1}]$. Therefore, C_{k+1} is a linear combination of $C_1, \ldots, C_k$ the augmented matrix does has a solution. In others words, the REF of does not introduce a new pivot when we row-reduce $[C_1 \cdots C_{k+1}]$. Applying the same sequence to EROs in the previous parts, we see that pivots are introduced in columns $1, 3$, and 5. Hence, the free variables are in

$$\boxed{C_2,\ C_4,\ C_6}.$$

Hence each of C_2, C_4, and C_6 is a linear combination of the preceding columns.

(c) Determine a basis for the column space of A.

Solution: From part (a), the row–echelon form of A has pivots in columns 1, 3, and 5. A basis for $\text{Col}(A)$ is given by:

$$\boxed{\{\, C_1,\ C_3,\ C_5 \,\}} = \left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 2 \\ 6 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \\ 5 \\ 14 \end{pmatrix} \right\}.$$

(d) Express column C_4 as a linear combinations of the columns in part (c).

Solution: We are finding solutions x, y, z such that

$$xC_1 + yC_3 + zC_5 = C_4$$

Setting up the augmented matrix gives:

$$(C_1, C_3, C_5|C_4) = \left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 2 & 3 & 7 & 7 \\ 1 & 2 & 5 & 5 \\ 3 & 6 & 14 & 15 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

We obtain the REF from the same sequence of EROs from the earlier part. Back-substitution from the echelon form gives:

$$\begin{aligned} z &= 0 && \text{Row 3} \\ y + z &= 3 && \text{Row 2} \\ x + y + 3z &= 2 && \text{Row 1.} \end{aligned}$$

Hence $z = 0$, $y = 3$, $x = 2 - 3 - 0 = -1$. Therefore,

$$C_4 = -C_1 + 3C_3 + 0C_5.$$

10 marks

2. A school teacher would like to perform a predictive analysis on the exam performance of the students in her class. She has taken a sample of students' scores for the following six subjects: **Mathematics**, **Science**, **English**, **Spanish**, **Computing**, and **Accounting**. Let $\vec{x} = (x_1, x_2, x_3, x_4, x_5, x_6)^\top$ denote a student's standardised scores for the six subjects in the given order.

Note: In the context of statistical inferences, *Standardisation* is the process of converting variables to a common scale (with mean 0 and standard deviation 1) for meaningful comparison of variables with different units or ranges.

The teacher would like to first perform *dimension reduction* to scale down the size of the data while retaining important information. She defines the linear map $T : \mathbb{R}^6 \to \mathbb{R}^4$, $\vec{x} \mapsto T(\vec{x})$, where the associated matrix $[T] = A$ is

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 2 \\ 1 & 1 & 2 & 2 & 1 & 3 \end{pmatrix}$$

The matrix A is specifically created for this context. For example, the first entry of $A\vec{x}$ can be interpreted as the *sum* or *average* performance of a student in all six subjects. A large value for the first entry of $A\vec{x}$ indicates that the student performs well overall, and vice versa. In the same way, the value for the second entry of $A\vec{x}$ can be interpreted as the *contrast* between STEM subjects and linguistic subjects, and so on.

Define the set of transformed scores to be $W = \{A\vec{x} \mid \vec{x} \in \mathbb{R}^6\}$.

(a) Show that W forms a subspace of $\mathbb{R}^4$ and hence determine the dimension of W.

Solution:

Since W is the span of the column vectors, Thm 3.34 tells us W forms a subspace of $\mathbb{R}^4$. Alternatively, since A is the associated matrix for a linear map with image W, it is a subspace by Prop 4.50.

Definition 4.42 tells us the dimension of W is simply the rank of the matrix A. By performing Gaussian elimination, we have

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 2 \\ 1 & 1 & 2 & 2 & 1 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & -2 & 0 & -1 \\ 0 & 0 & -1 & -1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 \end{pmatrix} \begin{matrix} \\ R_2 \to R_2 - R_1 \\ \\ R_4 \to R_4 - R_1 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & -2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 3 \end{pmatrix} \begin{matrix} \\ \\ R_3 \to 2R_3 - R_2 \\ R_4 \to 2R_4 + R_2 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & -2 & -2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \\ \\ \\ R_4 \to R_4 - 5R_3/3 \end{matrix}$$

The echelon form of A has 3 pivots, hence $\text{rank}(A) = 3$, which means $\dim(W) = 3$.

(b) A student's score is called *average* if their transformed scores are all zeros (as an element in W). Show that the set of vectors of $\mathbb{R}^6$ resulting in average scores forms a subspace. Determine (with justifications) the maximum number of linearly independent vectors in that subspace.

Solution: The set of vectors of $\mathbb{R}^6$ resulting in average scores is defined by the set $\left\{\vec{x} \in \mathbb{R}^6 \middle| A\vec{x} = \vec{0}\right\}$, and it is a subspace according to Theorem 3.32 (it is in fact called the kernel of A).
By the rank-nullity theorem (Thm 4.48), we know that $\text{rk}(A) + \dim \ker(A) = 6$. Since the $\text{rk}(A) = 3$ from part (a), we know immediately that $\dim \ker(A) = 3$. By Proposition 3.17, we know that the number of linearly independent vectors in $\ker(A)$ must be smaller than or equal to $\dim \ker(A)$. Thus, the maximum number of linearly independent vectors in $\ker(A)$ is 3.

(c) To improve the interpretability of the transformed scores in the set W, the teacher introduces a second linear map $S : \mathbb{R}^4 \to \mathbb{R}^4$, $\vec{y} \mapsto S(\vec{y})$, where the associated matrix $[S] = B$ is defined as

$$B = \begin{pmatrix} -1 & 2 & -1 & 1 \\ -1 & 0 & -1 & 1 \\ 4 & -2 & 2 & -2 \\ -3 & 2 & -1 & 3 \end{pmatrix}$$

We define the composite map $R = S \circ T$, and let the associated matrix $[R] = C$.
Compute the matrix C, and briefly explain what each entry in $C\vec{x}$ represents for a fixed but unknown set of scores $\vec{x} \in \mathbb{R}^6$.

Solution: Thm 4.12 tells us $C = [S \circ T] = [S] \times [T] = BA$. Hence,

$$C = \begin{pmatrix} -1 & 2 & -1 & 1 \\ -1 & 0 & -1 & 1 \\ 4 & -2 & 2 & -2 \\ -3 & 2 & -1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 2 \\ 1 & 1 & 2 & 2 & 1 & 3 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 2 & 0 & 0 & 2 & 0 \\ 0 & 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 4 \end{pmatrix}$$

The transformed scores can be written as

$$C\vec{x} = 2 \begin{pmatrix} x_1 + x_2 + x_5 \\ x_3 + x_4 \\ x_6 \\ x_1 + x_2 + x_3 + x_4 + x_5 + 2x_6 \end{pmatrix}$$

The first three entries of the vector $C\vec{x}$ is the *average* or *total* scores of a student in their STEM subjects, linguistic subjects, and accounting subject respectively. The fourth entry is dependent on the first three entries (based on our finding from part (a)) and hence is redundant. Note that the introduction of map S has improved the interpretability of the transformed scores, as a large value in each entry reflects the student's strength in the corresponding field/area.

(d) A student obtains a vector of subject scores given by $\vec{x}_J \in \mathbb{R}^6$ with transformed scores $\vec{z}_J = R(\vec{x}_J)$.

i. Starting with $\vec{z}_J$, determine (with justifications) whether it is possible to recover the transformed scores $\vec{y}_J = T(\vec{x}_J)$.

Solution: From $\vec{z}_J = R(\vec{x}_J) = (S \circ T)(\vec{x}_J) = S(T(\vec{x}_J)) = S(\vec{y}_J)$, we know that there is a unique vector $\vec{y}_J$ if the inverse of S exists, i.e., B^{-1} exists. By Thm 4.19 this is equivalent to showing that $\det(B) \neq 0$. To calculate $\det(B)$, we can first perform a series of row operations to reduce matrix B to echelon form.

$$B = \begin{pmatrix} -1 & 2 & -1 & 1 \\ -1 & 0 & -1 & 1 \\ 4 & -2 & 2 & -2 \\ -3 & 2 & -1 & 3 \end{pmatrix} \sim \begin{pmatrix} -1 & 2 & -1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 6 & -2 & 2 \\ 0 & -4 & 2 & 0 \end{pmatrix} \quad \begin{matrix} R_2 \to R_2 - R_1 \\ R_3 \to R_3 + 4R_1 \\ R_4 \to R_4 - 3R_1 \end{matrix}$$

$$\sim \begin{pmatrix} -1 & 2 & -1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 2 & 0 \end{pmatrix} \quad \begin{matrix} \\ \\ R_3 \to R_3 + 3R_2 \\ R_4 \to R_4 - 2R_2 \end{matrix}$$

$$\sim \begin{pmatrix} -1 & 2 & -1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix} \quad \begin{matrix} \\ \\ \\ R_4 \to R_4 + R_3 \end{matrix}$$

By Proposition 4.16, we note that

$$\det(B) = \det \begin{pmatrix} -1 & 2 & -1 & 1 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & -2 & 2 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

since all row operations are of the form $R_i \to R_i + \beta R_J$. Therefore, $\det(B) = -1(-2)(-2)(2) = -8$ by Thm 2.35. We have thus shown that $\det(B) \neq 0$. Hence the inverse exists and we can recover $\vec{y}_J$.

ii. Starting with $\vec{z}_J$, determine (with justifications) whether it is possible to recover the vector of original scores $\vec{x}_J$.

Solution: From the previous part, we've determined that we can obtain $\vec{y}_J$ from $\vec{z}_J$. So it remains to show whether we can recover $\vec{x}_J$ from $\vec{y}_J$. By Definition 4.17, the matrix $A = [T]$ is invertible if and only if T is a bijection. We will now show that T is not a bijection.
From (b), we saw deduced that $\dim \ker(A) = 3$. Hence there are non-zero vectors $\vec{u}$ for which $T(\vec{u}) = \vec{0}$, which means T fails to be injective and hence it cannot be bijective (by defn of bijection). Hence, A is not invertible, and it is not possible to recover the unique vector $\vec{x}_J$.

9 marks

3. Let I be the identity matrix and let B be a fixed but unknown 2×3 matrix. Define the following subset:

$$W_B = \{A \in M_{3\times 2} \mid BA = kI \text{ for some } k \in \mathbb{R}\}$$

(a) Show that W_B forms a subspace of $M_{3\times 2}$

Solution: To show that it is a subspace, we check the conditions (Definition 3.25):

- **Non-empty**: Let $U = 0$ (the zero matrix). Then $BU = 0I.$ so $0 \in W_B$. So the set is non-empty
- **Closure under addition**: Let $U, V \in W_B$. Then $BU = rI$ and $BV = sI$ for some $r, s \in \mathbb{R}$. We check that:

$$B(U+V) = BU + BV = rI + sI = (r+s)I$$

  Hence $B(U+V)$ is a scalar multiple of the identity. Hence $U+V \in W_B$
- **Closure under scalar multiplication**: Let $U \in W_B$ and $\alpha \in \mathbb{R}$. Check that:

$$B(\alpha U) = \alpha(BU) = (\alpha r)I$$

  Hence $B(\alpha U)$ is a scalar multiple of the identity. Hence $\alpha U \in W_B$

Since W_B meets all three criteria, it forms a subspace of $M_{3\times 2}$

(b) For the matrix $B = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \end{pmatrix}$, find a basis and hence the dimension of this space W_B:

Solution: Let $A \in W$ and label its elements as $A = \begin{pmatrix} a_1 & a_4 \\ a_2 & a_5 \\ a_3 & a_6 \end{pmatrix}$. Then we have:

$$\begin{aligned} BA &= \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} a_1 & a_4 \\ a_2 & a_5 \\ a_3 & a_6 \end{pmatrix} \\ &= \begin{pmatrix} a_1 + a_2 + a_3 & a_4 + a_5 + a_6 \\ 2a_1 + 3a_2 + a_3 & 2a_4 + 3a_5 + a_6 \end{pmatrix} \\ &= \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = kI \end{aligned}$$

Equating the entries gives the following system of linear equations:

$$\begin{cases} a_1 + a_2 + a_3 = k \\ 2a_1 + 3a_2 + a_3 = 0 \\ a_4 + a_5 + a_6 = 0 \\ 2a_4 + 3a_5 + a_6 = k \end{cases}$$

which can then be converted into an augmented matrix and solved:

$$\left(\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 0 & 0 & k \\ 2 & 3 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & 3 & 1 & k \end{array}\right) \sim \left(\begin{array}{cccccc|c} 1 & 1 & 1 & 0 & 0 & 0 & k \\ 0 & 1 & -1 & 0 & 0 & 0 & -2k \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & k \end{array}\right) \quad \begin{matrix} \\ R_2 \to R_2 - 2R_1 \\ \\ R_4 \to R_4 - 2R_3 \end{matrix}$$

From this, we can see two free variables. So let $a_3 = s, a_6 = t$, then back substitution gives:

$$\begin{aligned} a_5 - t &= k && \text{Row 4} \\ a_5 &= k + t \\ a_4 + a_5 + a_6 &= 0 && \text{Row 3} \\ a_4 + (k+t) + t &= 0 \\ a_4 &= -k - 2t \\ a_2 - s &= -2k && \text{Row 2} \\ a_2 &= -2k + s \\ a_1 + a_2 + a_3 &= k && \text{Row 1} \\ a_1 + (-2k + s) + s &= k \\ a_1 &= 3k - 2s \end{aligned}$$

Combining this information, we have every element $A \in W_B$:

$$\begin{aligned} A &= \begin{pmatrix} a_1 & a_4 \\ a_2 & a_5 \\ a_3 & a_6 \end{pmatrix} \\ &= \begin{pmatrix} 3k - 2s & -k - 2t \\ -2k + s & k + t \\ s & t \end{pmatrix} \\ &= s \begin{pmatrix} -2 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix} + t \begin{pmatrix} 0 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} + k \begin{pmatrix} 3 & -1 \\ -2 & 1 \\ 0 & 0 \end{pmatrix} \end{aligned}$$

Define the set

$$\mathcal{B} = \left\{ \begin{pmatrix} -2 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & -1 \\ -2 & 1 \\ 0 & 0 \end{pmatrix} \right\}$$

We claim that $\mathcal{B}$ is a basis for W_B.

- In our construction of $\mathcal{B}$, we show that $W_B = \text{span}(\mathcal{B})$.
- To show linear independence, we solve the equation:

$$s \begin{pmatrix} -2 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix} + t \begin{pmatrix} 0 & -2 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} + k \begin{pmatrix} 3 & -1 \\ -2 & 1 \\ 0 & 0 \end{pmatrix} = 0$$

  Comparing a_3 and a_6 shows that any solution must have $s = 0$ and $t = 0$. Then this forces the only solution of $k = 0$. Hence $s = t = k = 0$ is the only solution and hence $\mathcal{B}$ is a linearly independent set.

Thus $\mathcal{B}$ forms a basis of W_B and $\dim(W) = 3$.

12 marks

4. Let $\mathcal{P}_n(\mathbb{R})$ be the space of all polynomials of degree at most n, where an identified basis is $\{1, x, x^2, \ldots, x^n\}$. Let $\mathcal{F}(\mathbb{R})$ be the set space of all single variable functions. Define the following map $\mathcal{G} : \mathcal{P}_n(\mathbb{R}) \to \mathcal{F}(\mathbb{R})$, $f \mapsto \mathcal{G}(f)$ by

$$\mathcal{G}(f(x)) = \lim_{M\to\infty} \int_0^M e^{\omega x} f(x) dx \qquad \omega < 0.$$

You can assume without proof that the limit of the definite integral exists. For simplicity, we can write $\mathcal{G}(f(x)) = F(\omega)$ for some function F, since the mapped function is now expressed in terms of ω instead of x.

(a) Evaluate $\mathcal{G}(1), \mathcal{G}(x)$, and $\mathcal{G}(x^2)$.

Hint: Integration by parts might be useful.

Solution:

We first note that $\lim_{M\to\infty} M^k e^{\omega M} = 0$ for all integers $k \geq 0$ (this can be shown using the l'hopital's rule).

$$\mathcal{G}(1) = \lim_{M\to\infty} \int_0^M e^{\omega x} dx = \lim_{M\to\infty} \left[\frac{e^{\omega x}}{\omega}\right]_0^M = -\frac{1}{\omega}$$

$$\begin{aligned}
\mathcal{G}(x) &= \lim_{M\to\infty} \int_0^M x e^{\omega x} dx \\
&= \lim_{M\to\infty} \left\{ \left[\frac{x e^{\omega x}}{\omega}\right]_0^M - \int_0^M \frac{e^{\omega x}}{\omega} dx \right\} \\
&= \lim_{M\to\infty} \left[\frac{x e^{\omega x}}{\omega}\right]_0^M - \frac{1}{\omega} \lim_{M\to\infty} \int_0^M e^{\omega x} dx \\
&= 0 - \frac{1}{\omega} \cdot \left(-\frac{1}{\omega}\right) = \frac{1}{\omega^2}
\end{aligned}$$

$$\begin{aligned}
\mathcal{G}(x^2) &= \lim_{M\to\infty} \int_0^M x^2 e^{\omega x} dx \\
&= \lim_{M\to\infty} \left\{ \left[\frac{x^2 e^{\omega x}}{\omega}\right]_0^M - \int_0^M \frac{2x e^{\omega x}}{\omega} dx \right\} \\
&= \lim_{M\to\infty} \left[\frac{x^2 e^{\omega x}}{\omega}\right]_0^M - \frac{2}{\omega} \lim_{M\to\infty} \int_0^M x e^{\omega x} dx \\
&= 0 - \frac{2}{\omega} \cdot \left(\frac{1}{\omega^2}\right) = -\frac{2}{\omega^3}
\end{aligned}$$

(b) Prove that $\mathcal{G}$ is a linear map.

Solution: From the definition of linear maps, we have for any two functions $f, g \in$

$\mathcal{P}_n(\mathbb{R})$, we have

$$\begin{aligned}\mathcal{G}(f(x)+g(x)) &= \lim_{M\to\infty}\int_0^M e^{\omega x}[f(x)+g(x)]dx \\ &= \lim_{M\to\infty}\left[\int_0^M e^{\omega x}f(x)dx + \int_0^M e^{\omega x}g(x)dx\right]\end{aligned}$$

Since $\lim_{M\to\infty}\int_0^M e^{\omega x}f(x)dx$ and $\lim_{M\to\infty}\int_0^M e^{\omega x}g(x)dx$ exist, we have that

$$\begin{aligned}\mathcal{G}(f(x)+g(x)) &= \lim_{M\to\infty}\int_0^M e^{\omega x}f(x)dx + \lim_{M\to\infty}\int_0^M e^{\omega x}g(x)dx \\ &= \mathcal{G}(f(x)) + \mathcal{G}(g(x))\end{aligned}$$

Moreover, for any $\lambda \in \mathbb{R}$, we have

$$\begin{aligned}\mathcal{G}(\lambda f(x)) &= \lim_{M\to\infty}\int_0^M \lambda e^{\omega x}f(x)dx \\ &= \lambda\cdot\lim_{M\to\infty}\int_0^M e^{\omega x}f(x)dx \\ &= \lambda\mathcal{G}(f(x))\end{aligned}$$

Hence, $\mathcal{G}$ is a linear map.

(c) Consider a vector $f(x) \in \mathcal{P}_n(\mathbb{R})$. Denote the first derivative of $f(x)$ with respect to x as $f'(x) = \frac{d}{dx}f(x)$. Show that

$$\mathcal{G}(f'(x)) = -[f(0) + \omega F(\omega)].$$

Solution: By definition,

$$\begin{aligned}\mathcal{G}(f'(x)) &= \lim_{M\to\infty}\int_0^M e^{\omega x}f'(x)dx \\ &= \lim_{M\to\infty}\left\{[e^{\omega x}f(x)]_0^M - \int_0^M \omega e^{\omega x}f(x)dx\right\} \\ &= \lim_{M\to\infty}\left\{e^{\omega M}f(M) - f(0) - \omega\int_0^M e^{\omega x}f(x)dx\right\} \\ &= \lim_{M\to\infty} e^{\omega M}f(M) - \lim_{M\to\infty} f(0) - \omega\lim_{M\to\infty}\int_0^M e^{\omega x}f(x)dx \\ &= -f(0) - \omega F(\omega)\end{aligned}$$

since $\lim_{M\to\infty} e^{\omega M}f(M) = 0$.

(d) Prove by induction, that $\mathcal{G}(x^n) = n!/(-\omega)^{n+1}$ for integers $n \geq 0$.

Solution: To show by induction, we need two things:

- Base case: For $n = 0$, we have $\mathcal{G}(x^0) = \mathcal{G}(x^0) = \frac{1}{-\omega} = \frac{0!}{-\omega^{0+1}}$ - Solution from (a). So the base case holds.
- Suppose that $\mathcal{G}(x^n) = n!/(-\omega)^{n+1}$ is true for some $n \geq 0$, then

$$\begin{aligned}
\mathcal{G}(x^{n+1}) &= \lim_{M\to\infty} \int_0^M x^{n+1} e^{\omega x} dx \\
&= \lim_{M\to\infty} \left\{ \left[\frac{x^{n+1} e^{\omega x}}{\omega} \right]_0^M - \int_0^M \frac{(n+1)x^n e^{\omega x}}{\omega} dx \right\} \\
&= \lim_{M\to\infty} \left[\frac{x^{n+1} e^{\omega x}}{\omega} \right]_0^M - \frac{n+1}{\omega} \lim_{M\to\infty} \int_0^M x^n e^{\omega x} dx \\
&= 0 - \frac{n+1}{\omega} \cdot \left(\frac{n!}{(-\omega)^{n+1}} \right) \\
&= -\frac{(n+1)!}{(-\omega)^{n+2}}
\end{aligned}$$

Thus by induction, $\mathcal{G}(x^n) = n!/(-\omega)^{n+1}$ is true for all natural numbers n (in fact, it is also true for $n = 0$).

(e) Define $\mathcal{B} = \{1, x, x^2, \ldots, x^n\}$ and $\mathcal{C} = \{\mathcal{G}(1), \mathcal{G}(x), \mathcal{G}(x^2), \ldots, \mathcal{G}(x^n)\}$. Show that $\mathcal{G} : \text{span}(\mathcal{B}) \to \text{span}(\mathcal{C})$ is bijective. That is, prove that $\mathcal{G}$ is an isomorphism between the two sets. Hence, deduce that $\mathcal{G}^{-1}$ exists and is unique.

Hint: Since $\mathcal{G}$ is a linear map and preserves linear combinations, it is sufficient to show that the map between basis elements are bijective. That is, the bijection holds between the sets $\{1, x, x^2, \ldots, x^n\}$ and $\{\mathcal{G}(1), \mathcal{G}(x), \mathcal{G}(x^2), \ldots, \mathcal{G}(x^n)\}$.

Solution: To show $\mathcal{G}$ is bijective, we need to show that it is injective and surjective.

- Surjectivity follows from the definition of the two sets. Every vector in $\mathcal{C}$ is the image of some vector in $\mathcal{B}$ under the linear map $\mathcal{G}$.
- For injectivity, we want to show that $\mathcal{G}(x^p) = \mathcal{G}(x^q)$ implies that $p = q$. We need

$$\begin{aligned}
\int_0^\infty x^p e^{\omega x} dx &= \int_0^\infty x^q e^{\omega x} dx \\
\frac{p!}{(-\omega)^{p+1}} &= \frac{q!}{(-\omega)^{q+1}}
\end{aligned}$$

to be true for all choices of $\omega < 0$. We can choose $\omega = -1$ so that $p! = q!$. Note that $p! = q!$ does not imply $p = q$ since we could have the one exception case $0! = 1!$. If we can rule out the case $p = 0, q = 1$ (or $p = 1, q = 0$) then we can show that $\mathcal{G}(x^p) = \mathcal{G}(x^q) \implies p = q$.

Suppose (without loss of generality) that $p = 0, q = 1$, then we have $1/(-\omega) = 1/\omega^2$ which is not true for all $\omega < 0$ (say, $\omega = -2$). Thus, we can rule out the case $p = 0, q = 1$ (and $p = 1, q = 0$) and conclude that $p = q$.

Since $\mathcal{G}$ is a bijection, Defn 4.17 tells us that $\mathcal{G}$ has an inverse.

(f) Let $a \in \mathbb{R}$ in an arbitrary constant. Apply the map $\mathcal{G}$ on both sides of the differential equation

$$f'(x) = f(x) + a(x-1)$$

Use your result in parts (c) and (d) to find the solution with boundary condition $f(0) = 0$.
Note: To obtain full credits, you must solve the differential equation using results from the previous parts.

Solution: Applying $\mathcal{G}$ to both sides, we have

$$\begin{aligned} \mathcal{G}(f'(x)) &= \mathcal{G}[f(x) + a(x-1)] \\ \mathcal{G}(f'(x)) &= \mathcal{G}(f(x)) + a\mathcal{G}(x) - a\mathcal{G}(1) && \text{linearity} \\ -f(0) - \omega F(\omega) &= F(\omega) + \frac{a}{\omega^2} + \frac{a}{\omega} \end{aligned}$$

using the results from part (c). Since $f(0) = 0$, we can rewrite the equation as

$$\begin{aligned} -(\omega+1)F(\omega) &= \frac{a(\omega+1)}{\omega^2} \\ F(\omega) &= -\frac{a}{\omega^2} \end{aligned}$$

From part (d), we know that $\mathcal{G}^{-1}$ exists, hence we can find the solution $f(x)$ by applying $\mathcal{G}^{-1}$ on both sides of the equation

$$\begin{aligned} \mathcal{G}^{-1}(F(\omega)) &= \mathcal{G}^{-1}\left(-\frac{a}{\omega^2}\right) \\ \mathcal{G}^{-1}(F(\omega)) &= -a \cdot \mathcal{G}^{-1}\left(\frac{1}{\omega^2}\right) \\ f(x) &= -ax \end{aligned}$$